

RESEARCH INTERESTS

Geospatial Data Science
 Earth Observation
 Climate & Disaster Risk Assessment
 Earthquake Risk & Loss Modelling
 Physical Risk Modelling
 Exposure & Vulnerability Modelling
 Spatiotemporal Analysis
 Graph Neural Networks
 Probabilistic Deep Learning
 Environmental Data Science
 Multi-Hazard Risk Assessment
 Urban Analytics

TECHNICAL SKILLS

Python • R • MATLAB • PyTorch
 QGIS • Google Earth Engine • GDAL
 Jupyter • Git/GitHub • SLURM/HPC
 Weights & Biases • Tableau • Jekyll
 GitHub Pages • Mapbox • MapLibre

EDUCATION**UNIVERSITY OF CAMBRIDGE**

PH.D. ARTIFICIAL INTELLIGENCE FOR
 THE STUDY OF ENVIRONMENTAL RISKS
 M.RES. ENVIRONMENTAL DATA SCIENCE
 Oct. 2022 - Sep. 2026 | Cambridge, UK
Dissertation: Understanding Regional Dynamic
 Physical Risks: Advanced Geospatiotemporal
 Modelling of Exposure & Vulnerability

STANFORD UNIVERSITY

M.A. PUBLIC POLICY AND
 M.S. CIVIL & ENVIRONMENTAL ENG'G
 EXECUTIVE EDUCATION CERTIFICATE
 IN ENTREPRENEURSHIP & INNOVATION
 Sep. 2019 - Jun. 2022 | Stanford, CA, USA
Knight-Hennessy Scholar

UNIVERSITY OF THE PHILIPPINES

B.S. CIVIL ENGINEERING
 Jun. 2013 - Jan. 2018 | Los Baños, PH

TOKYO METROPOLITAN UNIVERSITY

EXCHANGE PROGRAM CERTIFICATE,
 INTEGRATED RIVER ENGINEERING WORK
 Oct. - Nov. 2017 | Tokyo, JP

PEER-REVIEWED WORKS

Nature Communications, **AAAI** Social Impact,
ICML ML for Remote Sensing, Environmental
 Research Letters, Progress in Disaster Science,
 Sustainable Cities and Society (*in review*),
 Science of Remote Sensing (*in review*), World
 Conf. Earthquake Engineering (**WCCE**),
 Int'l Disaster Risk Conf. (**IDRC**)

SCIENTIFIC REVIEWER

AAAI AI for Social Impact, **ICLR** ML for Remote
 Sensing, **UNDRR-ISC** Hazard Info Profiles,
Deep Learning Indaba, Cambridge J. of Sci. &
 Policy, Stanford J. of Sci., Tech., & Society

EXPERIENCE (8 1/2 years of research dev't, project mgt, & risk consulting)**UNIVERSITY OF CAMBRIDGE** | DOCTORAL RESEARCHER, RISK & RESILIENCE

Oct. 2022 - present | Cambridge, UK | 4+ years

- Improved the spatiotemporal resolution of physical exposure models across 44+ Least Developed Countries by 5x finer using geospatial AI on time-series satellite imagery.
- Forged a university-industry partnership in the Philippines to deliver city-scale multi-hazard urban risk knowledge products.

GERMAN AEROSPACE CENTER (DLR) | GUEST SCIENTIST, DIGITAL GREENTALENTS

Apr. - Jun. 2026 | Munich, Germany | 3 months

- Processed 28 GB of census data (543M+ rows) to produce the first nationwide maps of building typology at national, provincial, city, and barangay scale.

GERMAN AEROSPACE CENTER (DLR) | GUEST SCIENTIST, HELMHOLTZ AI

Jul. - Sep. 2024 | Munich, Germany | 3 months

- Designed a deep learning framework leveraging satellite imagery & census constraints to improve spatial disaggregation for building exposure and typology mapping.

EARTHQUAKES & MEGACITIES INITIATIVE (EMI) | URBAN RESILIENCE FELLOW

Mar. - Aug. 2022 | Quezon City, Philippines | 6 months

- Developed large-scale seismic risk assessment tools covering 400,000 buildings & 3.2M population under a scenario "Big One" M7.2 earthquake.
- Co-authored a climate and disaster risk assessment report for the Quezon City Government, a knowledge product covering earthquake, flood, and landslide hazards and identifying hotspot barangays to guide local resilience planning and resource allocation.

STANFORD UNIVERSITY, URBAN RESILIENCE INITIATIVE (SURI) | RESEARCHER

Dec. 2021 - May 2022 | Stanford, CA, USA | 6 months

- Conducted a probabilistic seismic risk assessment of 23 cities & municipalities in the Greater Metro Manila Area, Philippines, and benchmarked their disaster risk management capacities against the country's competitiveness index to evaluate policy priorities.

STANFORD LAND, BUILDINGS, & REAL ESTATE (LBRE) | SEISMIC RISK CONSULTANT

Mar. 2021 - Mar. 2022 | Stanford, CA, USA | 1 year

- Quantified earthquake-induced financial losses and business interruption risks for 750 buildings under seismic scenarios up to magnitude 7.6, and developed a stakeholder-facing Tableau dashboard to support institutional decision-making.

FM GLOBAL, ENG'G & RESEARCH | STRUCTURES & NATURAL HAZARDS RESEARCHER

Jun. - Sep. 2021 | Norwood, MA, USA | 4 months

- Expanded the firm's library of earthquake design hazard maps by more than 246%, from 130 to 320 maps (190 new maps) across over 130 countries.

STANFORD UNIVERSITY, STRUCTURES AS SENSORS LAB | RESEARCHER

Jun. 2020 - Jun. 2021 | Stanford, CA, USA | 1 year

- Implemented a Bayesian causal inference framework leveraging satellite imagery to enhance post-earthquake USGS landslide, liquefaction, & building damage models by 15x.

ARUP | STRUCTURAL DESIGN ENGINEER

Jul. 2018 - Jul. 2019 | Metro Manila, Philippines | 1 year

- Conducted post-earthquake assessment, performance- and code-based structural peer reviews for multiple 41-50 story high-rise developments.

AYALA GROUP | PLANNING & CONTROL ENGINEERING INTERN

May. - Jul. 2017 | Bonifacio Global City, Philippines | 3 months

- Built a VBA-Excel system tracking deliverables across 100+ concurrent projects, improving project management and monitoring efficiency.

ENGINEERING CONGRESS | COMMITTEE HEAD

Jun. 2016 - Jun. 2017 | Los Baños, Philippines | 1 year

- Developed event invitation and registration strategies that increased participation fivefold (300 to 1,500 attendees) for major engineering seminars and quiz competitions.